Hepa-Twin

A digital twin for early warning of decompensation in liver cirrhosis

Digital Twin Challenge 2026 | Happiest Health

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All data synthetic. Decision support only. Not for clinical use.

The problem: patients get worse between visits

Three challenges

- Cirrhosis is common in India: alcohol, fatty liver (MASLD), hepatitis B and C
- Many patients are diagnosed late
- Many live far from a liver specialist and visit only every month or two
- Between visits, the doctor knows nothing

Why it is dangerous

- A stable patient can suddenly decompensate:
- confusion, fluid in the belly, bleeding, serious infection, emergency admission
- Our question: can we spot warning signs days earlier, using data a phone or fitness band already records?

Warning signs a wearable can see

1. Flipped sleep cycle

- Awake at night, sleepy by day: early hepatic encephalopathy

2. Falling activity

- Fewer steps as the patient gets weaker

3. Rising resting heart rate

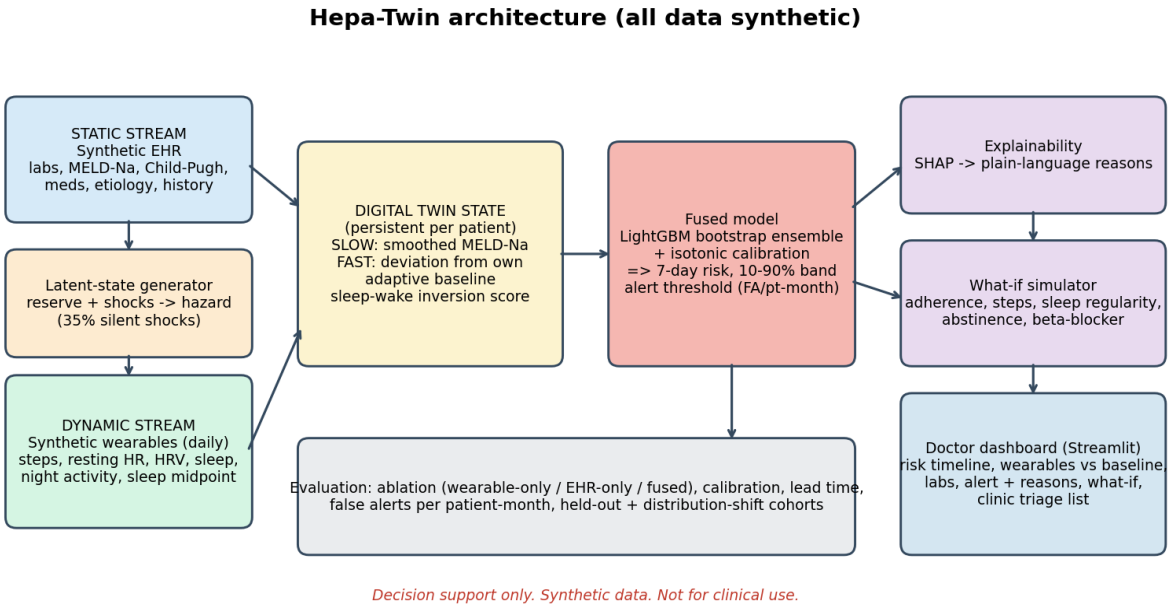
- Infection, bleeding or dehydration

4. Falling HRV

- Body's stress control weakening

Beta-blockers keep the heart rate low and hide signal 3, so the model must know who takes them.

The idea: a digital twin of the patient



- Slow data: medical record, labs, medicines, genetic marker
- Fast data: daily steps, heart rate, HRV, sleep
- Twin: slow severity plus fast deviation from the patient's OWN normal
- Output: 7-day risk, reasons, what-if, dashboard

The data: all synthetic, built honestly

What I generated

- 1,200 patients, 150 days each
- 700 train, 250 test, 250 harder shift group
- No Synthea cirrhosis module and no open cirrhosis + wearable dataset, so a custom generator

How events happen

- Hidden liver reserve declines slowly
- Random shocks build over 3-10 days
- Events are drawn by chance from hidden state, not copied from wearables
- 35% of shocks are silent

Realistic mess

- Days the band is not worn
- Weekend dips, hospital gaps
- Beta-blocker effect
- Shift group: more noise, 50% silent shocks

No real patient data is used anywhere (DPDP Act and HIPAA aware).

How the twin and features work

Twin state

- Learns each patient's normal in the first 7 days
- Updates that normal slowly, and only on normal days, so a decline is not absorbed
- Fast part: smoothed deviation from own normal
- Slow part: smoothed MELD-Na severity

87 features, one question

- 59 wearable features: averages, trends, deviation, sleep-wake inversion score
- 28 record features: labs, medicines, history, genetic marker
- Question: will an event happen in the next 7 days?
- The 7 days after an event are excluded (hospital stay)

The model: fused, calibrated, explainable

Prediction

- LightGBM, team of 6 models
- Split by patient, never by row
- Calibrated: 30% risk means about 30%
- Disagreement between models gives the uncertainty band
- Alerts tuned for at most 1 false alert per patient-month

Explain and simulate

- SHAP converted to plain-language reasons
- What-if: medicine adherence, steps, sleep regularity, alcohol, beta-blocker
- What-if shows learned patterns, not proof of cause

Results

Model	AUROC test	AUPRC test	AUROC shift
Wearable-only	0.724	0.205	0.657
EHR-only	0.746	0.206	0.717
Fused (EHR + wearable)	**0.776**	**0.263**	**0.717**

0.78
AUROC, final model

64%
events caught in advance

4 days
median warning

0.65
false alerts per patient-month

Fusing both streams beats either alone on the main test.

- Harder shift group: fused only ties EHR-only (0.717); false alerts rise to 1.83 per patient-month
- About a third of events missed, partly by design (silent shocks)
- Plain accuracy not reported: events are rare (about 7% of days)

Live demo: what the doctor sees

1. Clinic overview

- Alerts today, highest risk, average risk
- Needs-attention cards: red, amber, green
- 30-day trend and new alerts
- Day slider moves time

2. Patient twin

- Risk timeline with uncertainty band
- Wearables vs own baseline
- Alert with plain-language reasons
- Reveal ground truth toggle

3. What-if and triage

- Sliders re-score the patient instantly
- Triage list sorted by risk: who to call first
- About page lists limitations

Limitations, next steps

Limitations

- All data synthetic: method shown, not clinically validated
- Circularity risk reduced, not removed
- Medical constants are assumptions
- What-if is association; real-time is a replay

Next steps

- Test on real anonymised data
- Re-tune alerts per population
- Review with liver specialists
- Live wearable feeds, sleep stages

Hepa-Twin: slow hospital data + fast wearable data = a personal twin that warns a week ahead, explains why, and lets the doctor test what-ifs.

Thank You